Original Paper

Ophthalmic Research

Ophthalmic Res 2006;38:255–260
DOI: 10.1159/000094834

Received: October 25, 2005
Accepted: January 27, 2006
Published online: July 28, 2006

Combined Proton Beam Radiotherapy and Transpupillary Thermotherapy for Large Uveal Melanomas: A Randomized Study of 151 Patients

L. Desjardins L. Lumbroso-Le Rouic C. Levy-Gabriel R. Dendale S. Delacroix C. Nauraye M. Estève C. Plancher B. Asselain

Institut Curie, Paris, France

Key Words
Proton beam therapy · Uveal melanoma · Transpupillary thermotherapy · Glaucoma

Abstract
Introduction: Exudation from the tumour scar and glaucoma can be major problems after proton beam irradiation of uveal melanoma and can sometimes lead to secondary enucleation. We conducted a randomized study to determine whether systematic transpupillary thermotherapy (TTT) after proton beam radiotherapy could have a beneficial effect. ***Patients and Method:*** Between February 1999 and April 2003, all the patients treated by proton beam radiotherapy for uveal melanomas ≥7 mm thick or ≥15 mm in diameter were included in this study after giving their informed consent. One half of the patients received proton beam radiotherapy alone (60 Gy in 4 fractions) and the other half received the same dose of proton beam radiotherapy followed by TTT at 1, 6 and 12 months. All the information concerning the initial tumour parameters, treatments and follow-up was recorded and a statistical analysis was performed. ***Results:*** We randomized 151 patients. The median follow-up was 38 months. The 2 groups of patients were similar in terms of age, gender and tumour characteristics. The patients treated with TTT showed a greater reduction of tumour thickness ($p = 0.06$), less retinal detachment at the latest follow-up ($p = 0.14$) and a lower secondary enucleation rate ($p = 0.02$). ***Discussion:*** The present study is the first randomized analysis to demonstrate a significant decrease in the secondary enucleation rate in patients treated with TTT after proton beam radiotherapy. Further studies should be performed to determine whether TTT could be beneficial to smaller tumours and to define its optimal dose.

Introduction

There is a high rate of ocular complications after proton beam radiotherapy for large uveal melanomas. Severe glaucoma can sometimes require secondary enucleation. Our results for the treatment of uveal melanomas at the Orsay Proton Therapy Centre have been published in previous articles [1, 2]. In a series of 756 patients with tumours >15 mm in diameter and/or 7–12 mm thick and a follow-up of 5 years, the 5-year actuarial rates of glaucoma, retinal detachment and secondary enucleation were of 52, 27 and 13%, respectively. Studies conducted in Boston and Villigen have also demonstrated that, during proton beam radiotherapy of choroidal melanoma, a larger tumour volume is significantly correlated with an in-

KARGER
Fax +41 61 306 12 34
E-Mail karger@karger.ch
www.karger.com

0030–3747/06/0385–0255$23.50/0

Accessible online at:
www.karger.com/ore

Laurence Desjardins
Institut Curie
26, rue d'Ulm
FR–75005 Paris (France)
Tel. +33 1 44 32 46 03, Fax +33 1 53 10 40 18, E-Mail laurence.desjardins@curie.net

creased frequency of exudative, inflammatory and glaucomatous complications [3–5].

In isolated cases, we have obtained a reduction of exudative phenomena and glaucoma by treating the tumour scar with transpupillary thermotherapy (TTT). We therefore conducted a randomized study to determine whether the systematic addition of TTT after proton beam radiotherapy could have a beneficial effect.

Patients and Method

Between February 1999 and April 2003, patients treated by proton beam radiotherapy for uveal melanomas ≥7 mm thick or ≥15 mm in diameter were included in a randomized protocol comparing proton beam radiotherapy alone to proton beam radiotherapy followed by TTT. The protocol was proposed to all patients whose tumours fulfilled the size criteria and who did not have exclusion criteria, but some patients refused to be included in the study.

With a β-risk of 0.10 and an α-risk of 0.05, we estimated that 152 patients would be needed, i.e. 76 patients in each arm, to demonstrate a 25% reduction of the glaucoma rate (reduction from 25 to 50% with a power of 90%).

Proton beam radiotherapy was performed according to the usual technique, i.e. localizing tantalum clips were inserted at the Curie Institute under peribulbar or general anaesthesia during a 48-hour hospitalization. The tumour was located by transillumination or with a Schepens ophthalmoscope for amelanotic tumours. The distances measured in the operating room were compared to the data obtained from angiogrammes and fundus photographs. All the data were entered into the Eye Plane software, allowing 3-dimensional reconstruction of the eye and the tumour. The treatment planning obtained defined the ideal eye position during treatment. The patient positioning at the Orsay Proton Therapy Centre was ensured by a computer-controlled robotized chair with a precision of 0.1 mm. The patients were immobilized by a system comprising a mouthpiece and a mask. The chair was moved into the correct position by 2 orthogonal X-ray tubes detecting the tantalum clips. The eye movements during the treatment were monitored by a small camera connected to a light pen. A dose of cobalt-60 Gray equivalents was delivered in 4 fractions of 15 Gy on 4 days. The irradiation field included a safety margin of 2.5 mm around the tumour.

For the TTT a diode laser emitting in the red range at a wavelength of 810 nm was used. This laser is absorbed by pigmented tissues and induces necrosis of melanoma cells to a depth of 3 or 4 mm [6]. This effect is obtained by avoiding an excessively high energy, which would induce a photocoagulation effect with immediate blanching [7]. We used an energy allowing the progressive modification of the colour of the melanoma, which turned greyish.

One half of the patients included in the protocol received proton beam radiotherapy alone and the other half received proton beam radiotherapy followed by TTT at 1, 6 and 12 months. The inclusion criteria were a tumour diameter of ≥15 mm and/or a tumour thickness of ≥7 mm. Non-inclusion criteria were the presence of metastases, pre-existing glaucoma and opaque media preventing TTT (cataract, vitreous haemorrhage). All the patients received detailed information about the protocol and gave written informed consent.

The TTT was performed as an outpatient procedure, usually without local anaesthesia (the laser treatment being performed after the proton beam one, the treatment of necrotic tissue was not painful). We used many spots to cover the entire surface of the tumour with a total duration of 3 min. The slit-lamp delivery system was applied in all cases. However, TTT was not performed in the cases of recent vitreous haemorrhage or massive exudative detachment, and the patients were reviewed 6 months later. Some patients randomized to the laser group but with persistent vitreous haemorrhage or detachment did not receive laser treatment. This will be shown in the results. In some patients with an anterior tumour the anterior part of the tumour could not be reached and therefore was not treated.

All the patients were subjected to ophthalmological examination and liver ultrasound every 6 months. All the data concerning initial tumour characteristics and treatment were recorded. The ocular functional assessment and the thickness of the tumour scar were recorded annually. This assessment included visual acuity, presence or absence of palpebral sequelae, keratitis, cataract, glaucoma, retinal detachment, optic neuropathy or maculopathy, and inflammation. The patient's general health and the presence of local recurrence, secondary enucleation or metastases were also recorded. The statistical analysis was performed by the log-rank test.

Results

The present study included 151 patients, 78 males and 73 females with a median age of 59 years (range 22–88 years), corresponding to 68 right and 83 left eyes. The median follow-up was 38 months. The tumour was situated anterior to the equator in 10 patients, at the equator in 103 and posterior to the equator in 38. An initial retinal detachment was present in 65 patients, 33 presented ciliary body involvement and 9 optic disc involvement. The median initial tumour diameter was 17.5 mm (range 8–23.3 mm), and the median initial tumour thickness was 7.5 mm (range 2.4–11.8 mm). The median initial visual acuity was 20/60 (range 20/400–20/20). The initial assessment comprised angiography in 142 cases and ultrasound in 151. The initial tumour characteristics were comparable in the 2 groups (table 1).

The median characteristics of laser spots for patients treated by TTT were diameter 2 mm, spot intensity 500 mW and duration of each spot 3 min. Seven patients in the proton-beam-radiotherapy-only group received an application of TTT following the development of a complication (particularly massive exudation from the tumour scar or appearance of glaucoma). Nine patients in the proton-beam-radiotherapy-and-TTT group did not

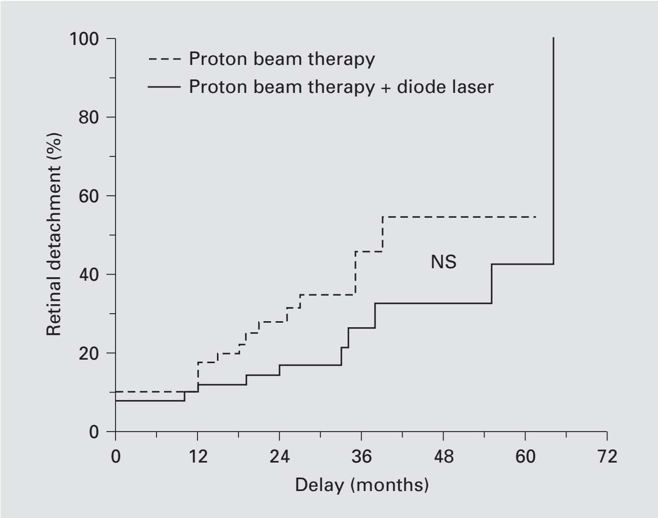

Fig. 1. Retinal detachment and group of treatment.

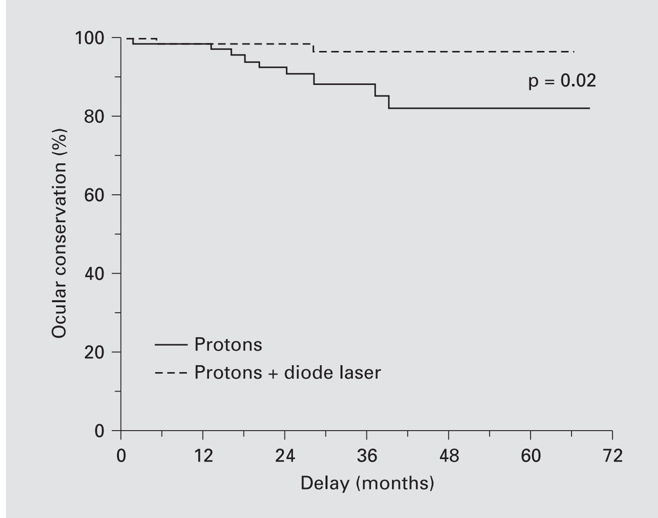

Fig. 2. Survival with ocular conservation and group of treatment.

Table 1. Initial tumour characteristics in the 2 groups

	Proton beam alone	Proton beam + TTT
Gender		
Male	45	33
Female	30	43
Age, years	56	59
Mean tumour diameter, mm	17.6	17.6
Mean tumour thickness, mm	7	7.6
Site of tumour		
Anterior	3	7
At the equator	54	49
Posterior	18	20
Initial retinal detachment	31	34
Ciliary body involvement	16	17
Tumour touching the optic disc	3	6

receive TTT because of retinal detachment or vitreous haemorrhage. The statistical analysis was performed according to the initial randomization, as it is the rule in randomized studies.

For patients who received TTT, the median intervals after proton beam radiotherapy were 37 days for the first TTT session, 6 months for the second and 12 months for the third. At the last follow-up, 126 patients were alive and 25 had died (23 from metastases and 2 from unrelated diseases). Of the surviving patients, 116 were in good health, 9 were treated for metastases and 1 was lost to follow-up. A total of 34 patients developed metastases.

No statistically significant difference was observed between the 2 groups in terms of cataracts, maculopathy, papillopathy and glaucoma. In the proton-beam-radiotherapy-only group 41 patients developed glaucoma, and the mean peak intraocular pressure recorded during treatment was 34.5 mm Hg. In the TTT group 35 patients developed glaucoma, and the mean peak intraocular pressure during treatment was 31 mm Hg. These figures are not significantly different, but the analysis of the course of glaucoma in the 2 groups showed that the patients who had received TTT presented less severe glaucoma, with intraocular pressure that tended to return to normal after several months of treatment. A tendency towards a reduction of the number of retinal detachments over time was observed in the TTT group (not significant; $p = 0.14$, fig. 1). A more marked reduction of tumour thickness was observed in the TTT group ($p = 0.06$), and the secondary enucleation rate was significantly lower in the TTT group ($p = 0.02$, fig. 2).

Discussion

After proton beam radiotherapy for a large uveal melanoma, the reduction of the tumour volume occurs progressively, and scarred pigmented tissue with a volume slightly smaller than that of the initial tumour often persists, but can nevertheless represent a fairly large mass. After 12–18 months, this scar may produce exudates that can spread away from the irradiated zone and sometimes induce exudative retinal detachment. Irradiation-induced tumour necrosis can lead to the release of lysosomal enzymes responsible for extensive necrosis [8, 9].

Uveitis and severely raised intraocular pressure refractory to medical treatment may be observed simultaneously. It is often too late to treat the scar by laser thermotherapy at this stage due to synechiae and lens opacification, which is why some eyes then become painful due to raised intraocular pressure and must be enucleated even in the absence of a tumour recurrence [10].

In some of the glaucomas observed after proton beam therapy there is a neovascularization of the iris, whereas in others the mechanism of glaucoma is not clear.

There are some uncertainties concerning the cause for neovascular glaucoma in patients treated for uveal melanoma with proton beams. With proton beam therapy, the area of ischaemic retina around the tumour is smaller than with plaque brachytherapy and is not likely to be the cause for neovascular glaucoma. However, retinal ischaemia can be worse if the retina is detached.

In large tumours the volume of the anterior segment receiving a significant amount of radiation is important and could therefore be the origin of neovascularization. On the other hand, it was recently shown by Boyd et al. [10] that irradiation induces iris atrophy and that the irradiated iris is resistant to neovascularization.

Uveal melanoma has been shown to secrete angiogenesis factors (vascular endothelial growth factor) [11]. We wonder whether the tumour scar could also secrete vascular endothelial growth factor or other mediators.

We also observed glaucoma in patients who have large tumour scars without any evidence of iris neovascularization. The question as to what role the tumour scar plays in these cases has not been answered.

In a retrospective study of 2,241 patients treated at the Curie Institute by iodine-125 plaque or proton beam radiotherapy, the secondary enucleation rate for complications with a median follow-up of 72 months was 6.5%, and the significant risk factor on multivariate analysis was the tumour diameter ($p < 0.0001$; EVER abstract, October 2004).

According to Egger et al. [12], the secondary enucleation rate after proton beam radiotherapy was 8.2% with a median follow-up of 44 months. The most significant prognostic factors were tumour volume and retinal detachment. Munzenrider et al. [13] reported a secondary enucleation rate of 5.7% after proton beam radiotherapy, and the most significant risk factors were tumour volume and ciliary body involvement. According to Egan et al. [14], the enucleation rate after proton beam radiotherapy was 6.4% with a median follow-up of 2.7 years, and Damato and Lecuona [15] reported a 5-year secondary enucleation rate of 11.1% after various conservative treatments (proton beam radiotherapy, plaque radiotherapy and surgical excision). The risk factors for secondary enucleation were a nasal tumour, optic disc involvement and tumour volume.

TTT has been used alone to treat small choroidal melanomas, but the local recurrence rate observed with this treatment was much higher than with radiotherapy, which limits its indications [16, 17]. Some authors use TTT as a complement to radiotherapy to improve local control, particularly with ruthenium, which emits less penetrating β-radiation than the low-energy γ-radiation of iodine-125 [18–21]. In this case, TTT is used to destroy the superior part of the tumour, while the part adjacent to the sclera is destroyed by irradiation. TTT has been used more recently to reduce exudative complications after proton beam radiotherapy. At the Curie Institute, we treated 2,200 patients with uveal melanoma by proton beam radiotherapy and observed a marked clinical improvement with the use of TTT in patients with severe exudates and exudative retinal detachment.

In 2003, Char et. al. [22] reported a more rapid resorption of the retinal detachment in a retrospective study of 11 patients treated by proton beam radiotherapy and TTT. More recently, Bechrakis et al. [23] described the toxicity associated with large scars after proton beam radiotherapy and proposed endoresection of the scars by vitrectomy.

In the present randomized study, the objective of TTT was to reduce the glaucoma rate, as we believe that large scars after proton beam radiotherapy are responsible for both exudation and secretion of chemical mediators that can induce necrosis of adjacent tissues, inflammation and raised intraocular pressure with or without neovascularization. TTT might be able to reduce this by destroying the tumour apex and maybe also by creating a dense adhesion between the tumour and the overlying retina.

The strengths of the study are that it is randomized and done with consecutive patients by a single institution.

Its weaknesses are that the protocol could not be followed in the case of opaque media at the time of laser exposure and also that some patients who should not have received laser therapy were still treated. Moreover, the present study does not appear to indicate any benefit in terms of glaucoma, but this could be due to the modalities of data collection. The presence or absence of glaucoma was recorded in the database during the ocular functional assessments performed at 1, 2, 3, 5 and 10 years. In order to assess the difference between the 2 groups of patients, the intraocular pressure and the duration of raised intraocular pressure, as well as the anti-glaucoma treatments used would need to be recorded. An individual review of the case files of our series revealed that the patients treated by TTT had lower intraocular-pressure figures and a shorter duration of raised intraocular pressure, which is why this patient group had a significantly lower secondary enucleation rate ($p = 0.02$). However, further studies based on precise data concerning glaucoma are necessary to analyse the results of TTT.

Furthermore, as TTT induces superficial necrosis of the tumour, it is not surprising that the group of patients treated with this modality presented a decreased tumour thickness. Serous retinal detachment tended to resolve more rapidly in the TTT group, but the difference between the 2 groups was not statistically significant. The TTT protocol used in this study was an empirical one, and further studies are necessary to determine whether TTT should be applied every 6 months or more frequently. This half-yearly protocol was selected as many patients live a long way from the Curie Institute and can only attend every 6 months. For some scars, it could be more effective to perform TTT more frequently to reduce the volume of toxic scar tissue more rapidly. Moreover, in all patients, it is essential to perform regular fluorescein angiography follow-up, treat any ischaemic territories by argon laser, operate on occlusive cataracts to allow the surveillance of the ocular fundus and laser therapy, and medically treat inflammation and glaucoma. Finally, the combination of TTT and proton beam radiotherapy may also be useful for smaller tumours, particularly to decrease exudation of scars situated adjacent to the macula.

Conclusion

Overall, this is the first randomized study that appears to demonstrate a benefit of TTT combined with proton beam radiotherapy in large tumours to allow long-term preservation of the eyeball. Many ophthalmologists use TTT combined with radiotherapy either to reduce the recurrence rate or to decrease exudation, but the results are only shown in retrospective non-randomized studies. The indications and benefits of TTT after proton beam therapy need to be further studied, and the optimal TTT protocol should be defined.

References

1 Desjardins L, Levy C, d'Hermies F, Frau E, Schlienger P, Habrand JL, Mammar H, Schwartz L, Mazam A, Delacroix S, Nauraye C, Ferrand R, Asselain B: Initial results of proton therapy in choroidal melanoma at the d'Orsey Center for Proton Therapy: the first 464 cases. Cancer Radiother 1997;1:222–226.

2 Desjardins L, Lumbroso L, Levy C, Mazal A, Delacroix S, Rosenwald JC, Dendale R, Plancher C, Asselain B: Treatment of uveal melanoma with iodine-125 plaques or proton beam therapy: indications and comparison of local recurrence rates. J Fr Ophtalmol 2003;26: 269–276.

3 Egger E, Zografos L, Munkel G, Bohringer T, Bercher L, Chamot L: Results of proton radiotherapy for uveal melanomas. Front Radiat Ther Oncol 1997;30:111–122.

4 Gragoudas ES, Seddon JM, Egan KM, Glynn R, Munzenrider JE, Austin-Seymour M, Goiten M, Verhey L, Urie M, Koehler A: Long-term results of proton beam irradiated uveal melanomas. Ophthalmology 1987;94:349–353.

5 Gragoudas ES, Li W, Goitein M, Lane AM, Munzenrider JE, Egan KM: Evidence-based estimates of outcome in patients irradiated for intraocular melanoma. Arch Ophthalmol 2002;120:1665–1671.

6 Oosterhuis JA, Journee-de Korver HG, Kakebeeke-Kemme HM, Bleeker JC: Transpupillary thermotherapy in choroidal melanomas. Arch Ophthalmol 1995;113:315–321.

7 Midena E, Pilotto E, de Belvis V, Zaltron S, Doro D, Segato T, et al: Choroidal vascular changes after transpupillary thermotherapy for choroidal melanoma. Ophthalmology 2003; 110:2216–2222.

8 Shields CL, Naseripour M, Cater J, Shields JA, Demirci H, Youseff A, Freire J: Plaque radiotherapy for large posterior uveal melanomas (≥8 mm thick) in 354 consecutive patients. Ophthalmology 2002;109:1838–1849.

9 Gunduz K, Shields CL, Shields JA, Cater J, Freire JE, Brady LW: Plaque radiotherapy of uveal melanoma with predominant ciliary body involvement. Arch Ophthalmol 1999; 117:170–177.

10 Boyd SR, Gittos A, Richter M, Hungerford JL, Errington RD, Cree IA: Proton beam therapy and iris neovascularisation in uveal melanoma. Eye 2005;16:16.

11 Boyd SR, Tan D, Bunce C, Gittos A, Neale MH, Hungerford JL, Charnock-Jones S, Cree IA: Vascular endothelial growth factor is elevated in ocular fluids of eyes harbouring uveal melanoma: identification of a potential therapeutic window. Br J Ophthalmol 2002;86: 448–452.

12 Egger E, Zografos L, Schalenbourg A, Beati D, Bohringer T, Chamot L, Goiten G: Eye retention after proton beam radiotherapy for uveal melanoma. Int J Radiat Oncol Biol Phys 2003; 55:867–880.

13 Munzenrider JE, Gragoudas ES, Seddon JM, Sisterson J, McNulty P, Birnbaum S, et al: Conservative treatment of uveal melanoma: probability of eye retention after proton treatment. Int J Radiat Oncol Biol Phys 1988;15: 553–558.

14 Egan KM, Gragoudas ES, Seddon JM, Glynn RJ, Munzenreider JE, Goitein M, Verhey L, Urie M, Koehler A: The risk of enucleation after proton beam irradiation of uveal melanoma. Ophthalmology 1989;96:1377–1382, discussion 1382–1383.

15 Damato B, Lecuona K: Conservation of eyes with choroidal melanoma by a multimodality approach to treatment: an audit of 1,632 patients. Ophthalmology 2004;111:977–983.

16 Harbour JW, Meredith TA, Thompson PA, Gordon ME: Transpupillary thermotherapy versus plaque radiotherapy for suspected choroidal melanomas. Ophthalmology 2003;110:2207–2214, discussion 2215.

17 Shields CL, Shields JA, Perez N, Singh AD, Cater J: Primary transpupillary thermotherapy for small choroidal melanoma in 256 consecutive cases: outcomes and limitations. Ophthalmology 2002;109:225–234.

18 Damato B, Patel I, Campbell IR, Mayles HM, Errington RD: Local tumor control after ^{106}Ru brachytherapy of choroidal melanoma. Int J Radiat Oncol Biol Phys 2005;20:20.

19 Bartlema YM, Oosterhuis JA, Journee-de Korver JG, Tjho-Heslinga RE, Keunen JE: Combined plaque radiotherapy and transpupillary thermotherapy in choroidal melanoma: 5 years' experience. Br J Ophthalmol 2003;87:1370–1373.

20 Shields CL, Cater J, Shields JA, Chao A, Krema H, Materin M, Brady LW: Combined plaque radiotherapy and transpupillary thermotherapy for choroidal melanoma: tumor control and treatment complications in 270 consecutive patients. Arch Ophthalmol 2002;120:933–940.

21 Stoffelns BM, Kutzner J, Schopfer K, Frising M: Prospective nonrandomised analysis of 'sandwich therapy' for malignant melanoma of the choroid. Klin Monatsbl Augenheilkd 2002;219:211–215.

22 Char DH, Bove R, Phillips TL: Laser and proton radiation to reduce uveal melanoma-associated exudative retinal detachments. Trans Am Ophthalmol Soc 2003;101:53–56, discussion 56–57.

23 Bechrakis NE, Hocht S, Martus P, Kreusel KM, Heese J, Foerster MH: Endoresection following proton beam irradiation of large uveal melanomas. Ophthalmologe 2004;101:370–376.